Excretory Products and Their Elimination

Introduction

1. Which of the following is an incorrect match of the organisms and their excretory wastes?
1) Reptiles, land snails and birds-Uricotelic
2) Insects, frog and bony fish-Ureotelic
3) Mammals, frog and shark—Ureotelic
4) Aquatic insects and aquatic amphibiansAmmonotelic

2. The most toxic excretory product is
1) Ammonia
2) Urea
3) Uric acid
4) CO_2

3. Which of the following is an incorrect statement?
1) Many terrestrial amphibians, mammals and marine fishes excrete urea.
2) Flame cells are the excretory structures in flatworms, rotifers, some annelids and cephalochordates.
3) Small amount of uric acid may be retained in the kidney matrix of some animals to maintain desired osmolarity.
4) The chief excretory waste of humans is synthesized by liver and eliminated by kidneys.

4. Given below are two statements.
Statement I: Ammonotelic animals excrete ammonia by diffusion across body surface or through gill surface (in fish) as ammonium ions.
Statement II: Kidneys do not play any significant role in excretion in ammonotelic animals.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both statements I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, but statement II is correct.
4) Both statements I and II are correct.

5. Match the following columns.

	Column I (Organism)		Column II (Excretory Structure)
1	Crab	(A)	Antennary gland or green gland
2	Housefly	(B)	Malpighian tubule
3	Human	(C)	Liver, kidney
4	Amphioxus	(D)	Protonephridia or flame cells

1) 1-(B), 2-(c), 3-(D), 4-(A)
2) 1-(A), 2–(B), 3–(C), 4-(D)
3) 1-(D), 2–(A), 3–(B), 4(C)
4) 1–(C), 2-(A), 3-(B), 4-(D)

6. Protonephridia are primarily concerned with
1) Ionic regulation
2) Fluid volume regulation
3) Osmoregulation
4) All of the above

7. Ammonia produced by metabolism is converted into urea in the

1) Lungs
2) Kidney
3) Liver
4) Spleen

8. Read the following statements and choose the correct option.

(i) Ammonia is the most toxic form and requires least amount of water for its elimination.
(ii) Kidneys do not play any significant role in removal of ammonia.
(iii) Terrestrial adaptation necessitated the production of most toxic nitrogenous wastes like urea and uric acid for conservation of water.
(iv) In most of the vertebrates, excretory structures are simple tubular forms.

How many statements are correct?

1) One
2) Two
3) Three
4) Four

Human Excretory System and Urine Formation

9. Which of the following statements is correct?

(i) Nitrogenous wastes are absorbed by passive transport.
(ii) Reabsorption of water occurs passively in the initial segment of nephron.
(iii) Conditional reabsorption of Na and water takes place in DCT.
(iv) DCT is capable of selective secretion of H^+, K^+, and NH_3 to maintain pH and Na^+-- K^+ balance in blood.

1) (i) and (ii)
2) (ii) and (iii)
3) (iv) and (v)
4) All of these

10. Which of the following is not considered as a component of tubular parts of nephron?

1) DCT
2) Loop of Henle
3) PCT
4) Glomerulus

11. Extension of cortex in between renal medullary pyramids are known as

1) Ducts of Bellini
2) Columns of Bertini
3) Collecting ducts
4) Renal fascia

12. Match the following columns and choose the correct option.

	Column I		Column II
1	Length of kidney	a	2-3 cm
2	Width of kidney	b	5-7 cm
3	Thickness of kidney	c	10-12 cm
4	Number of nephrons in each kidney	d	One million
		e	Ten million

1) 1-c, 2-b, 3-a, 4-d
2) 1-c, 2-a, 3-b, 4-e
3) 1-b, 2- a, 3-a, 4-d

4) 1-a, 2-b, 3-c, 4-e

13. Urine formation involves three main processes that takes place in different parts of the nephron in the sequence of

1) Glomerular filtration → Tubular secretion→ Reabsorption
2) Reabsorption → Tubular secretion → Glomerular filtration
3) Glomerular filtration→ Reabsorption→ Tubular secretion
4) Tubular secretion → Reabsorption → Glomerular filtration

14. Read the following statements and choose the correct option:

Statement I: Each nephron has two parts, i.e., glomerulus and renal tubule.

Statement II: Vasa recta is absent or highly reduced in juxtamedullary nephrons.

1) Both statements are correct.
2) Both statements are incorrect.
3) Only statement (I) is correct.
4) Only statement (II) is correct.

15. Peritubular capillaries of the nephron arise from

1) Vasa recta
2) Renal artery
3) Afferent arteriole
4) Efferent arteriole

16. Which of the following is found in the medullary pyramid of kidneys?

1) Loop of Henle
2) DCT
3) Malpighian corpuscle
4) PCT

17. Select the incorrect statement w.r.t. human excretory system:

1) Each kidney measures 10–12 cm in length.
2) The weight of each kidney is 70–100 g.
3) Hilum is a notch towards the centre of the inner concave surface of the kidney.
4) Kidneys are located between T_{12}-L_3 vertebra.

18. Ureter, blood vessels and nerves enter into the kidney through

1) Cortex
2) Medulla
3) Hilum
4) Columns of Bertini

19. Blood from the glomerulus is carried away by an

1) Afferent arteriole
2) Efferent arteriole
3) Renal artery
4) Renal pelvis

20. During urine formation, which of the following substances is not secreted by the tubular cells into the filtrate?

1) Hydrogen ion
2) Glucose
3) Ammonia
4) Potassium ion

Functions of Tubules and Mechanism of Concentration of the Filtrate

21. Urine produced by human kidneys is concentrated _____ times as compared to the initial filtrate formed.
1) Two
2) Four
3) Three
4) Five

22. The moving fluid is urine in all of the following ducts, except
1) Renal pelvis
2) Major calyx
3) Collecting duct
4) Loop of Henle

23. In the descending limb of the loop of Henle, the urine becomes _____ . Choose the option which correctly fills up the blanks.
1) Hypotonic
2) Hypertonic
3) Isotonic
4) All of these

24. Conditional reabsorption of Na+ and water takes place in __ A ___ in the presence of __ B __ Fill in the blanks.
1) A-PCT; B-ADH
2) A-DCT; B-Aldosterone
3) A-Loop of Henle; B-Aldosterone
4) A-Collecting duct; B-ADH

25. Which of the following parts of nephron is impermeable to water?
1) Collecting duct
2) PCT
3) Ascending limb of the loop of Henle
4) Descending limb of the loop of Henle

26. The osmolality of outer cortex in kidney is ___ while that of inner medulla is ____ . Fill in the blanks
1) 300 mOsm/L, 1200 mOsm/L
2) 1200 mOsm/L, 300 mOsm/L
3) 900 mOsm/L, 2000 mOsm/L
4) 1200 mOsm/L, 1500 mOsm/L

27. If the PCT lined by simple cuboidal brush-border epithelium is cut from a nephron, how will it effect the process of urine formation?
1) Urine will be more dilute.
2) Facultative reabsorption of Na^+ and water will not take place.
3) Selective secretion of H^+, K^+, NH_3 will not occur.
4) Counter current mechanism will stop, and filtrate will remain isotonic.

28. Counter current is formed by the
1) Flow of filtrate in the two limbs of Henle's loop in opposite directions
2) Flow of blood through the two limbs of vasa recta in the same direction
3) Flow of blood in the two limbs of Henle's loop in opposite directions
4) All of the above

29. 70-80 per cent of electrolytes are reabsorbed in

1) DCT
2) PCT
3) Ascending limb of Henle's loop
4) Descending limb of Henle's loop

30. Which part of nephron allows reabsorption of small amount of urea into medullary interstitium?
1) PCT
2) DCT
3) Collecting duct
4) Descending limb of Henle's loop

31. Removal of descending limb of Henle's loop from the nephron will result in
1) No change in quality and quantity of urine
2) No urine formation
3) More diluted urine
4) More concentrated urine

32. NaCl is transported by _____ which is exchanged with ____ of vasa recta.
1) Ascending limb of Henle's loop, Descending limb
2) Descending limb of Henle's loop, Ascending limb
3) Ascending limb of Henle's loop, Ascending limb
4) Descending limb of Henle's loop, Descending limb

33. Which of the following does not occur in response to release of angiotensin II?
1) Vasoconstriction
2) Increase in glomerular blood pressure
3) Release of aldosterone
4) Decreased reabsorption of Na" from renal tubules

34. If a person drinks very small amount of water, then after prolonged and strenuous exercise the body fluid will have
1) Antidiuretic hormone
2) Renin
3) Atrial natriuretic factor
4) Both (1) and (2)

35. What is the pH ofurine?
1) 6
2) 6.5
3) 7
4) 7.5

36. Which of the following does not occur in response to release of atrial natriuretic factor (ANF)?
1) Decreases blood pressure
2) Stimulates renin-angiotensin mechanism
3) Causes vasodilation
4) Increase the volume ofurine

37. Increase in blood volume and blood pressure stimulates certain specific cells of cardiac atria to secrete a peptide hormone called
1) Renin
2) Angiotensin
3) Aldosterone
4) Atrial natriuretice factor

38. If a person passes a very dilute urine but his blood glucose level is normal and experiences excessive thirst, then it may be due to

1) Fall of blood sugar level
2) Decrease in vasopressin release from neurohypophysis
3) Excessive release of insulin
4) Excessive release of glucagon

39. Which of the following stimulates adrenal cortex to release aldosterone and tackle the water stress condition?
1) Angiotensin II
2) Atrial natriuretic factor
3) Renin
4) Anti-diuretic hormone

40. Which of the following hormone is responsible for reabsorption of water from the distal parts of nephron?
1) Renin
2) Rennin
3) Vasopressin
4) Atrial natriuretic factor

41. Renin-Angiotensin mechanism secretes which of the following hormones?
1) Glucocorticoid
2) Mineralocorticoid
3) Both (1) and (2)
4) None of the above

42. Which of the following can increase blood pressure due to vasoconstrictor action?
1) Atrial natriuretic factor
2) Aldosterone
3) Vasopressin
4) Both (2) and (3)

43. Vasopressin responsible for reabsorption of water from DCT is synthesized by
1) Anterior pituitary
2) Posterior pituitary
3) Hypothalamus
4) Juxtaglomerular cells

44. Which of the following is correct w:2t. function of renin?
1) Reduce urine formation
2) Conversion of angiotensinogen into angiotensin I
3) Vasodilation
4) Formation of Graafian follicle

45. The process of release of urine is called
1) Micturition
2) Micturition reflex
3) Ultrafiltration
4) Both (1) and (2)

46. An increase in body fluid due to excess intake of water
1) Reduces the release of vasopressin
2) Increases glomerular blood flow and therefore GFR
3) Both (1) and (2)
4) Increases release of vasopressin

Role of Other Organs in Excretion and Disorders of Excretory System

47. Read the following statements and choose the correct option.
1) Our lungs remove significant quantities of water every day.
2) The sweat and sebaceous glands in the skin can eliminate certain substances through their secretions..
3) Small amounts of nitrogenous wastes could be eliminated through saliva too.
4) All statements are correct

48. Sebaceous glands secrete sebum containing
1) Waxes
2) Sterols
3) Hydrocarbons
4) All of these

49. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason(R).
Assertion (A): Secretion of sebaceous gland provides a protective oily covering for the skin.
Reason (R): Sebaceous glands eliminate certain substances like sterols, hydrocarbons and waxes through sebum.
In the light of the above statements, choose the correct answer from the options given below:
1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is not correct.
3) (A) is not correct but (R) is correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

50. Which of the following glands does not help in excretion?
1) Liver
2) Sweat glands
3) Pancreas
4) Both (1) and (2)

51. In which of the following organs, worn out RBCs degrade to form bile pigments like bilirubin and biliverdin?
1) Intestine
2) Liver
3) Lungs
4) Kidney

52. Ketone bodies are observed in urine of patients with
1) Starvation
2) Diabetes mellitus
3) Both (1) and (2)
4) Diabetes insipidus

53. Accumulation of urea in blood of patients suffering with renal failure is called
1) Cystitis
2) Haematuria
3) Uremia
4) Nephritis

54. The dialysing unit contains a coiled cellophane tube surrounded by a fluid (dialysing fluid) having the same composition as that of
1) Plasma

2) Nitrogenous wastes
3) Plasma except the nitrogenous wastes
4) Formed elements except the nitrogenous wastes

55. Match the following columns.

	Column I		Column II
A	Gout	(i)	Mass of crystallised salts within the kidney
B	Glycosuria	(ii)	Glomeruli inflammation
C	Renal calculi	(iii)	Uric acid accumulation in joints
D	Glomerulinephritis	(iv)	Glucose present in urine

1) A-i, B-iii, C-ii, D-iv
2) A-iv, B-iii, C-i, D-ii
3) A-iii, B-iv, C-i, D-ii
4) A-iii, B-ii, C-i, D-iv

56. Which one of the following is an abnormal constituent of urine?
1) Glucose
2) Na^+ ions
3) Creatinine
4) Hippuric acid

Answer Keys

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
2	1	3	4	2	4	3	1	4	4	2	1	3	3	4
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
1	2	3	2	2	2	4	2	2	3	1	3	1	2	3
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
3	1	4	4	1	2	4	2	1	3	2	4	3	2	1
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**				
3	4	4	4	3	2	3	3	3	3	1				